

Homework 1: Discrete State Dynamic Programming

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1 Model and household solvers: part (a)

I solve the household problem at fixed prices implied by $K^0 = 5$, with $\beta = 0.96$, $\sigma = 1.5$, labor supply one, efficiency states $(0.8, 1.2)$, and $P_{ss'} = 0.5$. With $(z, \alpha, \delta) = (1, 0.36, 0.1)$, the prices are $w^0 = 1.1423762763$ and $r^0 = 0.0285173311$. The Bellman and Euler derivations are in `manual/derivation.pdf`; numerical definitions follow `spec.md`.

The baseline uses $N = 1000$ uniform points on $[0, 20]$, zero initial values, and

$$d(x) = \max_m \frac{|\mathcal{T}(x)_m - x_m|}{|x_m| + 1} \leq 10^{-8}.$$

Counts include the stopping check; updates equal checks minus one.

Table 1: Baseline household solvers. Setup and solve times are separate.

Method	Checks	Inner steps	Setup (s)	Solve (s)	$d(x)$	Status
VFI	379	0	0.039	2.436	9.71806×10^{-9}	converged
Howard	15	0	0.036	0.702	6.01925×10^{-16}	converged
Modified Howard	15	700	0.033	0.132	1.65884×10^{-13}	converged
Fixed-step GD	20000	0	0.155	163.109	1.2576	unconverged
Adam	20000	0	0.145	159.890	0.125301	unconverged

The contraction calculation $\log(10^{-8})/\log(0.96) = 451.244$ predicts the scale of an eight-order error reduction; VFI takes 379 checks. The difference reflects the relative residual criterion, initial error, and realized contraction. Howard evaluates each improved policy exactly and terminates after few policy changes. Modified Howard uses $H = 50$ inner evaluations per update, accelerating convergence while avoiding a linear solve. Its 15 checks require 700 inner steps. I select modified Howard, the fastest passing solver in these runs.

Fixed-step GD uses $\eta = 5 \times 10^{-4}$, below the conservative local stability limit $2/(1 + \beta\sqrt{2000})^2 \simeq 0.00103623$; Adam uses learning rate 0.1. Both remain unconverged at 20,000 checks. For a fixed greedy policy, let $J = I - \beta Q_G$. The assignment's lower-bound benchmark is $\kappa(J^T J) \geq (1 - \beta)^{-2} = 625$.

Table 2: Local singular values and condition numbers, from `results/d.theory.json`.

Policy	$s_{\min}(J)$	$s_{\max}(J)$	$\kappa_2(J)$	$\kappa_2(J^T J)$
$x = 0$	0.00131756	30.3592	23042	5.30934×10^8
Howard solution	0.0199828	2.03095	101.635	10329.7

The severe conditioning at zero values is consistent with GD stalling. These fixed-policy diagnostics do not guarantee global convergence. The Howard time includes the first SciPy import, so timings describe the measured runs.

2 Invariant distribution: parts (b)–(d)

Using the same Howard policy, I solve $Q_G^T \pi = \pi$ with $\mathbf{1}^T \pi = 1$ by power iteration, inverse iteration, and a normalized equation system.

Table 3: Dense and CSR distribution methods. Times exclude common matrix construction/conversion; residuals are $\|Q_G^T \pi - \pi\|_\infty$.

Method	Dense (s)	CSR (s)	Dense residual	CSR residual
Power iteration	0.282998	0.006451	7.76899×10^{-13}	7.76906×10^{-13}
Inverse iteration	0.162382	0.008551	1.38778×10^{-17}	1.38778×10^{-17}
Modified equations	0.140231	0.007548	2.70617×10^{-16}	1.04083×10^{-17}

The largest pairwise differences are 5.49004×10^{-12} within dense, 5.48995×10^{-12} within CSR, and 5.49005×10^{-12} across formats. All pass the 10^{-8} agreement criterion and distribution checks. I select CSR power iteration: it is the fastest passing option and uses 212 updates. All subsequent experiments retain the chosen methods.

3 Asset range and grid: parts (e)–(g)

At $N = 1000$, support means asset nodes with marginal probability above 10^{-12} ; support share is their fraction of all nodes.

Table 4: Uniform-grid range trials. The accepted top-mass threshold is 10^{-12} .

$k_{\max}$	Top-node mass	Support endpoint	Support share
2	6.01277×10^{-5}	2.0000	100.0%
5	0	3.2182	63.4%
10	0	3.2132	32.2%
20	0	3.2633	16.4%
40	0	3.2432	8.2%

I reject $k_{\max} = 2$ and choose $k_{\max} = 5$, the smallest tested range with zero top mass and room above its support endpoint. Wider ranges reduce resolution for a fixed number of nodes.

On $[0, 5]$, I compare the uniform grid with the exp-log grid

$$k_n = \exp\left[\frac{n-1}{N-1} \log(1 + k_{\max})\right] - 1, \quad n = 1, \dots, N.$$

Table 5: Grid comparison at $N = 1000$; error statistics use positive-saving states.

Grid	Mean error	Maximum error	Weighted mean
Uniform	0.00120403	0.00412096	0.001472
Exp-log	0.00108912	0.00612422	8.04657×10^{-4}

I choose exp-log spacing for its lower mean and stationary-weighted errors, accepting its higher overall maximum. Both grids have zero top mass.

Table 6: Exp-log refinement on $[0, 5]$. Total numerical time excludes verification and plot export.

N	Mean assets	Mean error	Maximum error	Weighted mean	Time (s)
100	0.439733	0.0117694	0.0469941	0.00870041	0.688
500	0.450191	0.00225802	0.010252	0.00150289	0.086
1000	0.450854	0.00108912	0.00612422	8.04657×10^{-4}	0.195
2000	0.450965	5.63096×10^{-4}	0.00297408	3.95872×10^{-4}	0.718
5000	0.451125	2.23223×10^{-4}	0.00118456	1.55824×10^{-4}	4.363

I select $N = 5000$: it has the lowest tested errors and an acceptable runtime of 4.36 seconds. Mean assets stabilize near 0.45. They need not equal K^0 , because prices are fixed. At $N = 100$, one upper-bound choice is flagged, with top mass 2.67276×10^{-53} ; the finer grids have neither upper-bound choices nor top mass. Figures 2–6 plot both shocks at every N ; distribution curves show joint masses, not densities. My dated October 1–2 choices are in `log.md`.

4 Euler accuracy and tests: part (h)

For positive savings, define

$$c_{n,s}^{EE} = \left[\beta(1+r^0) \sum_{s'} P_{ss'} (c_{n,s,s'}^{next})^{-\sigma} \right]^{-1/\sigma}, \quad E_{n,s} = \left| 1 - \frac{c_{n,s}^{EE}}{c_{n,s}} \right|.$$

The equality applies off the borrowing constraint and artificial upper bound; discrete choices leave a residual. On $A = \{g_{n,s} > 0\}$, normalize stationary weights as $q_{n,s} = \pi_{n,s} / \sum_A \pi$. The weighted maximum is $\max_A(qE)$.

Table 7: Selected $N = 5000$ Euler errors. Support maximum restricts to positive-saving states with joint probability above 10^{-12} .

Mean	Maximum	Weighted mean	$\max(qE)$	Support maximum
2.23223×10^{-4}	0.00118456	1.55824×10^{-4}	6.44445×10^{-6}	9.66045×10^{-4}

Regressing log error on log maximum grid step gives slopes 1.01312 (mean) and 0.932503 (maximum). Figure 1 supports approximately first-order scaling over these grids; it does not establish an asymptotic theorem.

Table 8: Final-grid numerical checks. All meet their registered criteria.

Final-grid check	Measured value
Policy monotonicity violations	0
Minimum consumption	0.913901
Maximum budget discrepancy	0
Distribution sum	1.000000
Minimum probability	0
Stationarity residual	8.52166×10^{-13}
Top-node mass / upper-bound choices	0 / 0

The selected household solve uses 14 checks, 13 updates, and 650 inner steps, with $d(x) = 6.31419 \times 10^{-9}$; power iteration uses 168 updates. The agent’s final audit records 484 passes, zero failures, and zero pending checks. Its clean-copy reproduction matches the saved numerical outputs exactly.

5 Independent verification and AI use

The **2026-10-02 — Mishuk verification results** entry in `log.md` records two personal checks: the $N = 100$ manual-kernel comparison gives 492 checks in both implementations, maximum value difference 1.42×10^{-14} , and zero policy mismatches; the saved-output calculations give zero budget discrepancy, distribution sum 1.0, and Euler-error agreement within 2.22×10^{-16} . The October 1–2 Codex entries document its automated tests, experiments, and reproduction. Those agent measurements are distinct from my dated personal checks.

Codex (GPT-6; exact build unavailable) ran locally with workspace-restricted repository access. It implemented the solvers and tests, ran experiments, and generated the saved numerical results, original tables, and figures. ChatGPT assisted with the report-check prompt, drafting and shortening the report, compact table formatting, layout, and draft PDF previews. I supplied the inputs and manual reference, selected methods and grids, performed the recorded personal verification, reviewed the report, and compiled the current PDF.

A Required numerical figures

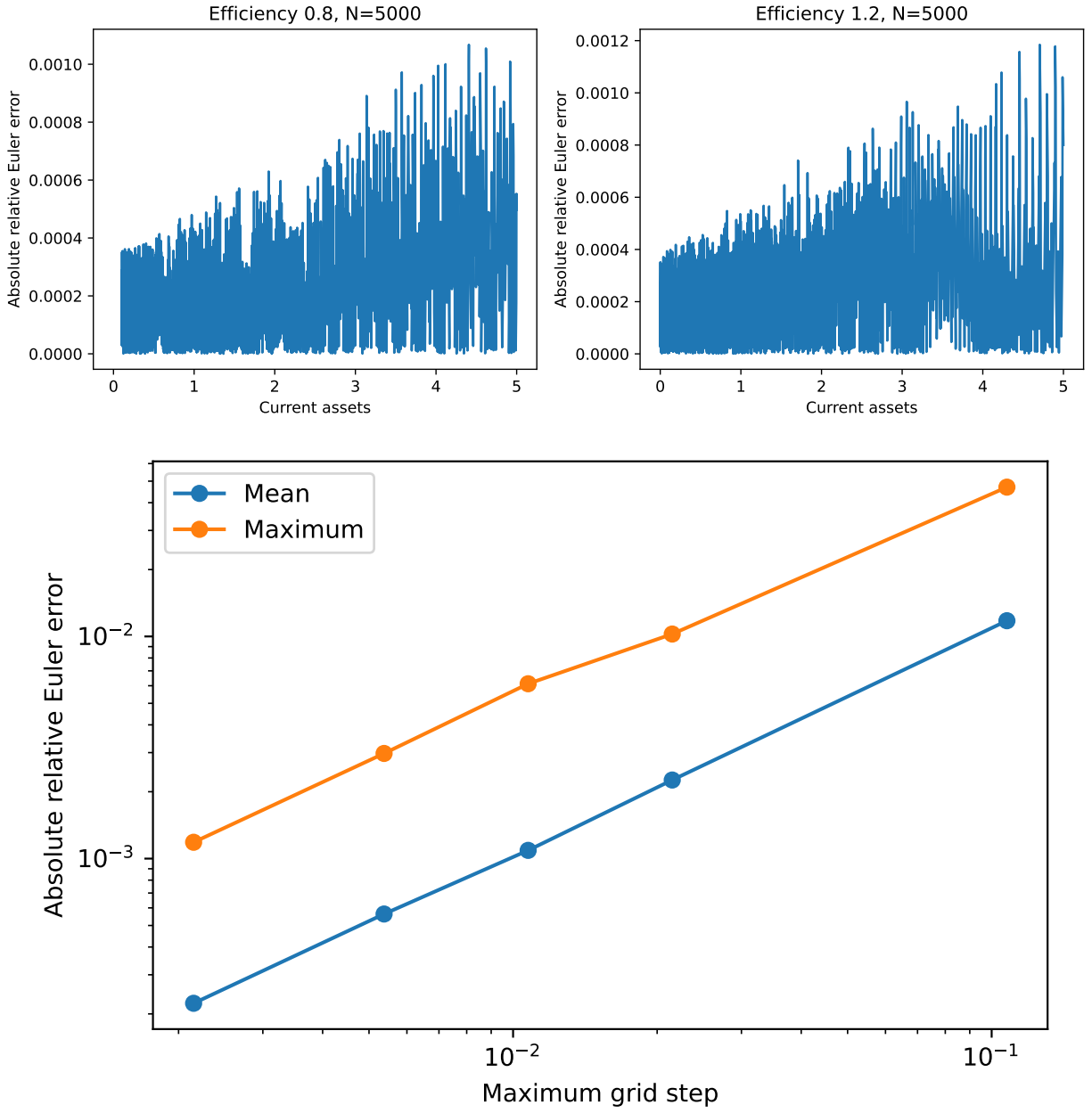


Figure 1: Selected-grid Euler residuals (top) and scaling across all five grids (bottom).

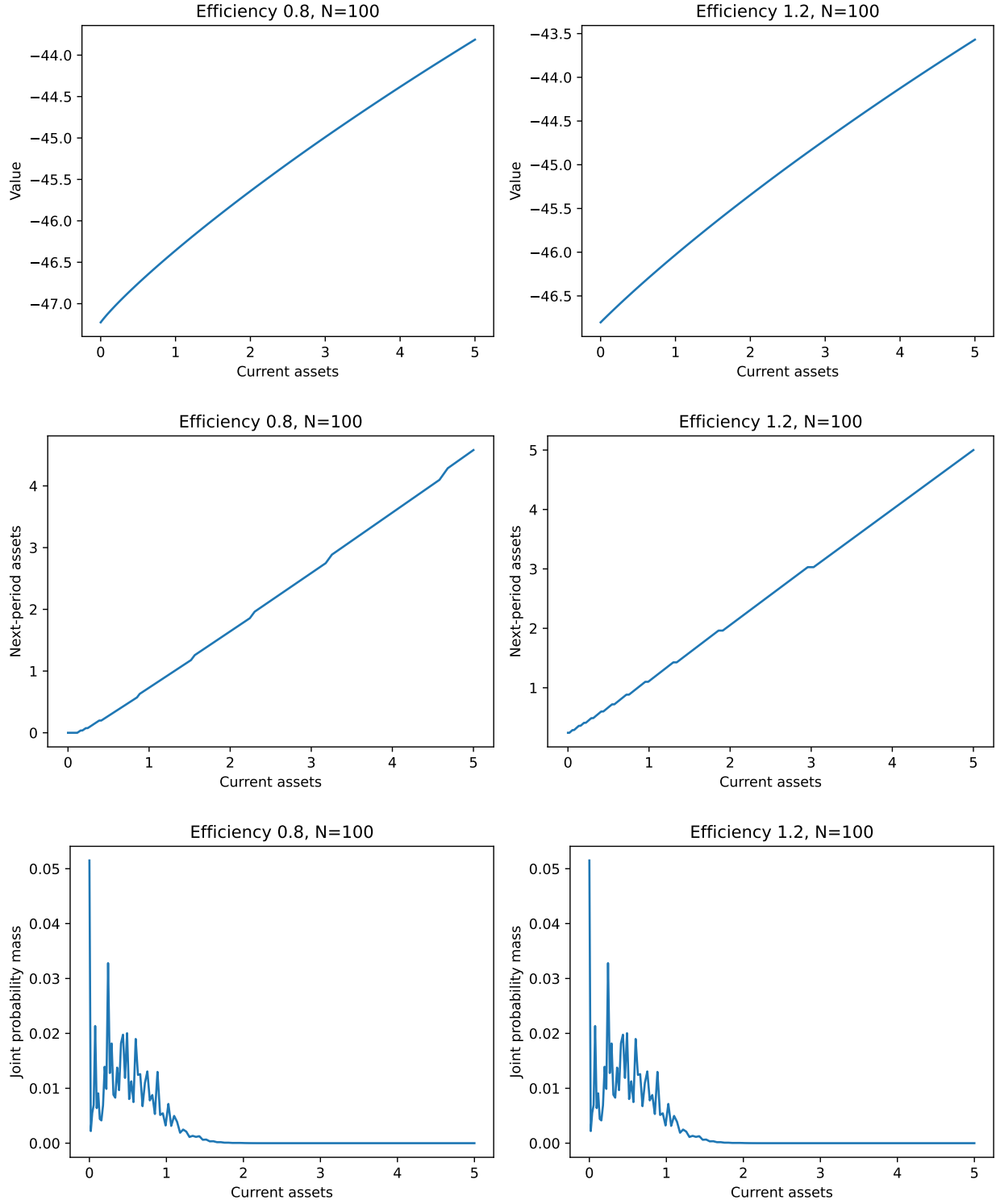


Figure 2: Values (top), savings policies (middle), and joint invariant probability masses (bottom) at $N = 100$, for both labor-efficiency states. Masses are probabilities at grid nodes, not densities.

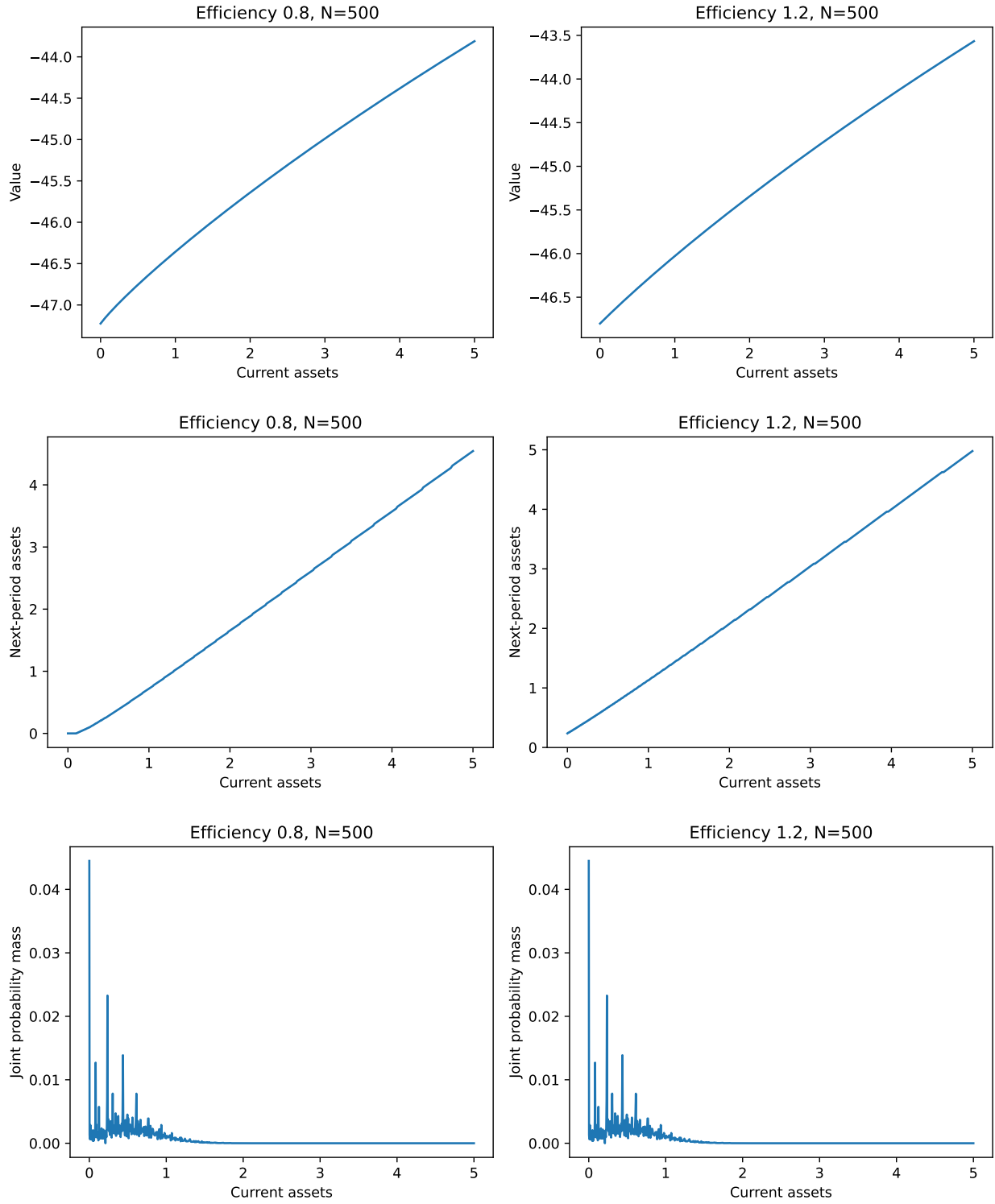


Figure 3: Values (top), savings policies (middle), and joint invariant probability masses (bottom) at $N = 500$, for both labor-efficiency states. Masses are probabilities at grid nodes, not densities.

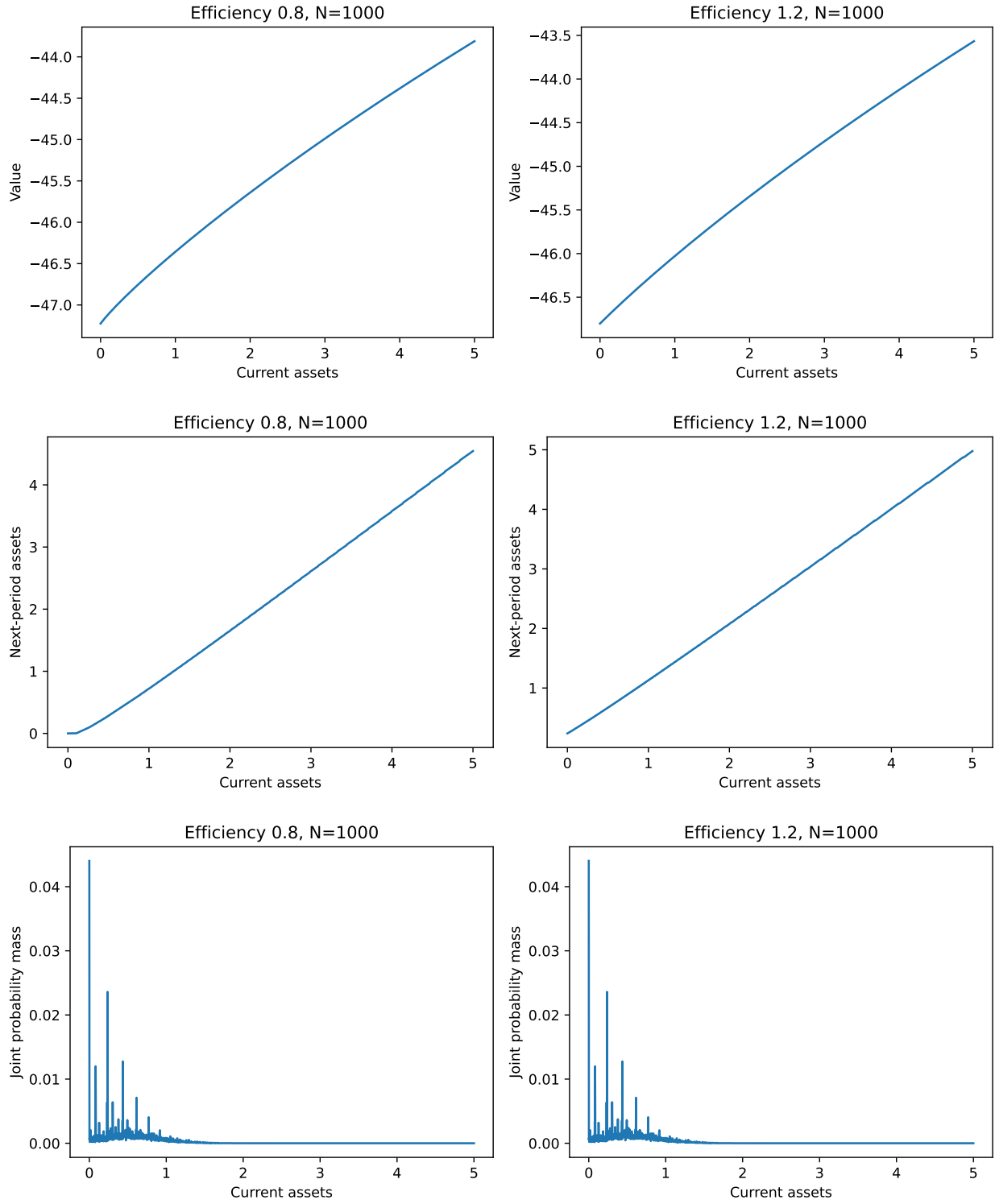


Figure 4: Values (top), savings policies (middle), and joint invariant probability masses (bottom) at $N = 1000$, for both labor-efficiency states. Masses are probabilities at grid nodes, not densities.

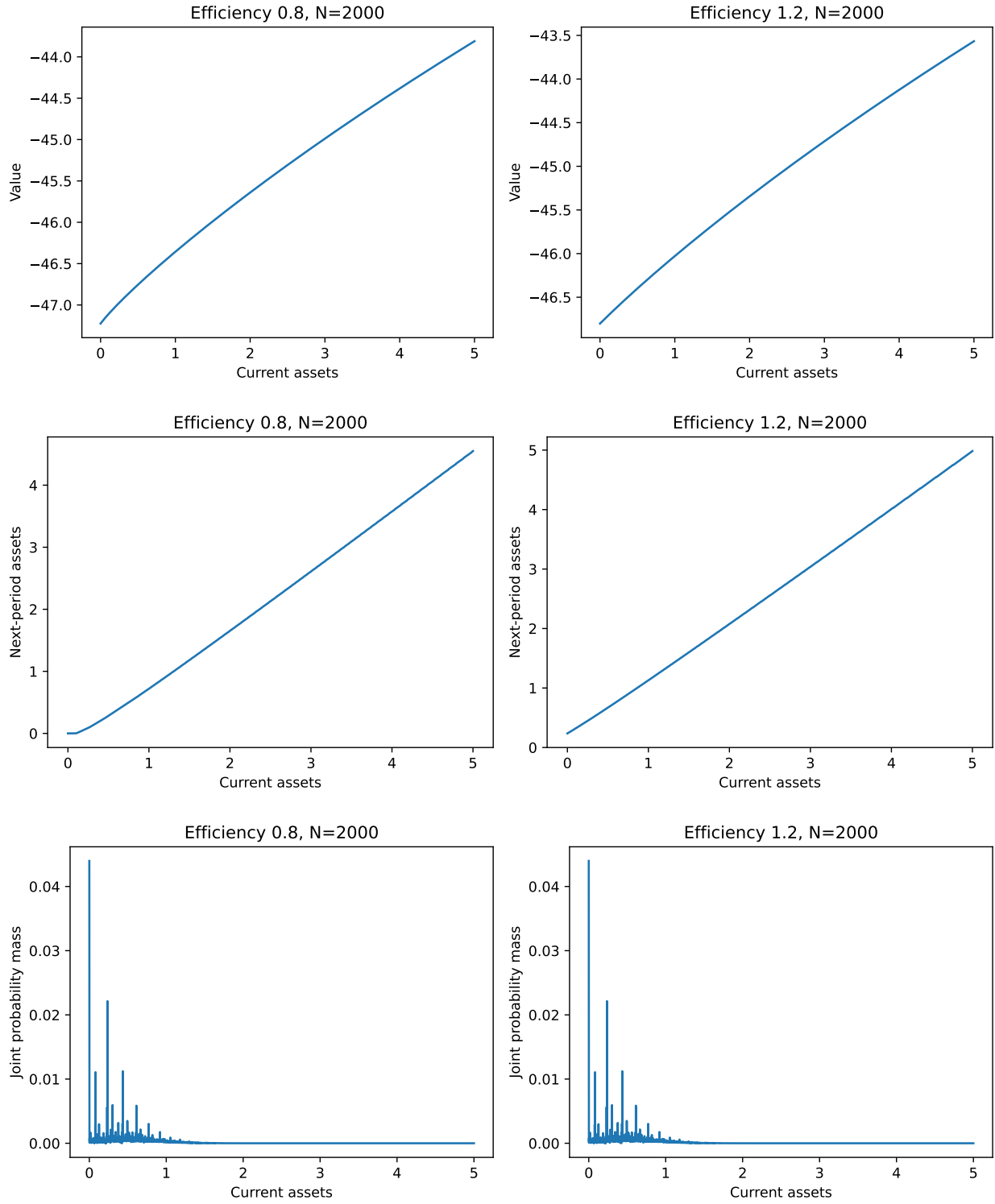


Figure 5: Values (top), savings policies (middle), and joint invariant probability masses (bottom) at $N = 2000$, for both labor-efficiency states. Masses are probabilities at grid nodes, not densities.

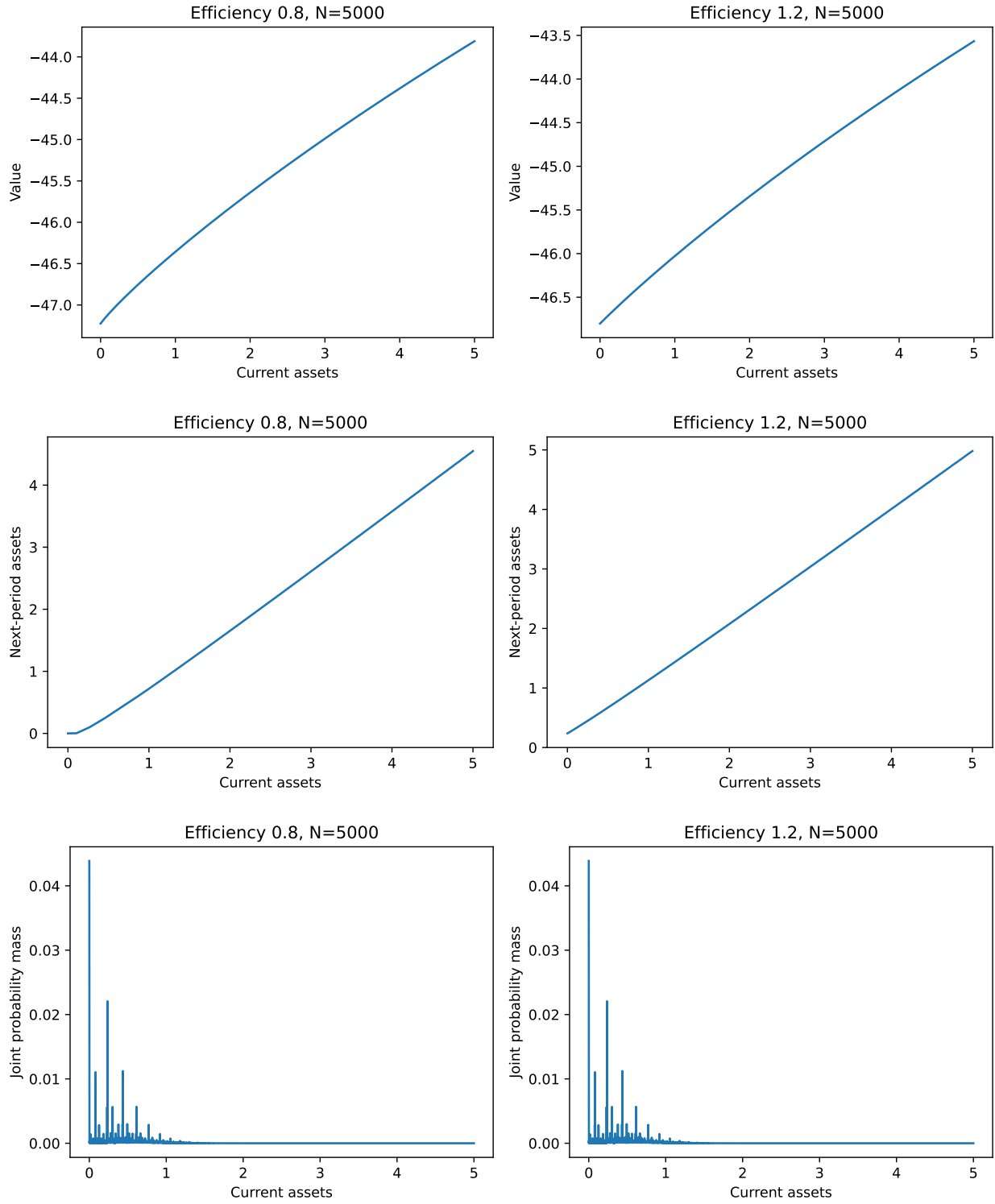


Figure 6: Values (top), savings policies (middle), and joint invariant probability masses (bottom) at $N = 5000$, for both labor-efficiency states. Masses are probabilities at grid nodes, not densities.